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The Seneca Polytechnic logo, featuring the word "Seneca" in a large, bold, red sans-serif font.

HOW FLIGHT PRICES VARY ACROSS AIRLINE, CLASSES, AND BOOKING WINDOWS IN INDIA –PART ONE: DATA VISUALIZATION

The Carleton University logo, consisting of the text "Carleton University" in a black serif font, with a red maple leaf icon to the right.The Tableau logo, featuring a cluster of colorful plus signs on the left and the word "tableau" in a blue sans-serif font on the right.

INTRODUCTION

- Airline ticket prices in India are widely depending on the factors like time of bookings, airlines, class, number of stops.
 - Understanding this patterns helps travellers to make decisions and airlines optimize pricing
 - Analysis contains over 300,000 flight records with information with airline, class, stops, days until departure, and price
 - Goal of this project to explore how different factors influence average flight prices in India, finding trends and provide actionable insights
 - Tableau will be used for data visualization and SAS will conduct for statistical analysis and regression modelling.
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WHERE DID I GOT THE DATASETS?

- Got datasets from Kaggle's source
- The datasets named is “*Flight Price Prediction*”

kaggle

PLAN FOR THIS PART (PART ONE)



In this personal project, the plan for this part one will be data visualization.



It will conduct in Tableau to create charts and dashboards.

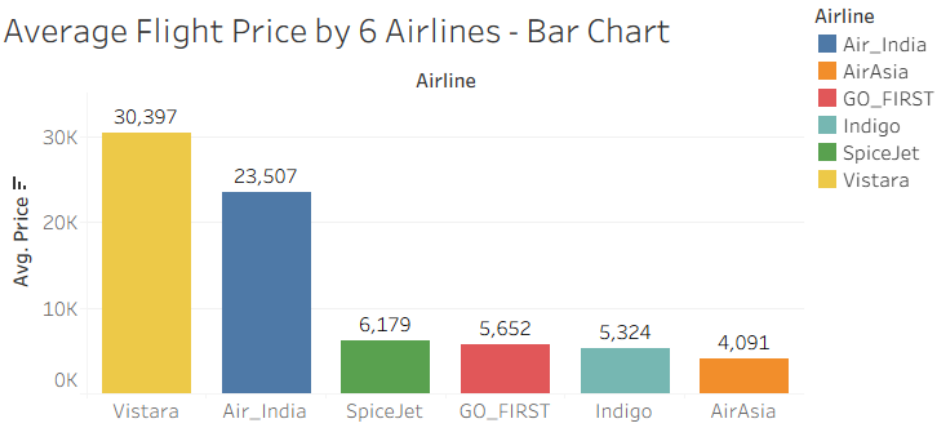


Will create another PowerPoint Slides for Part Two: Statistical Modelling and Regression Modelling in SAS.

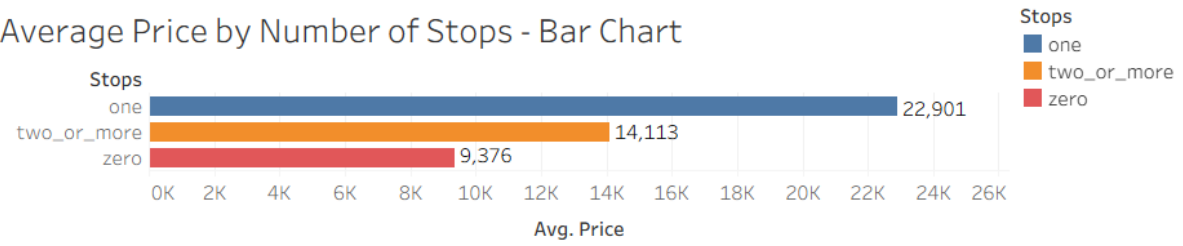
DASHBOARD OF TABLEAU

Average Price Based on the Factors - Dashboards

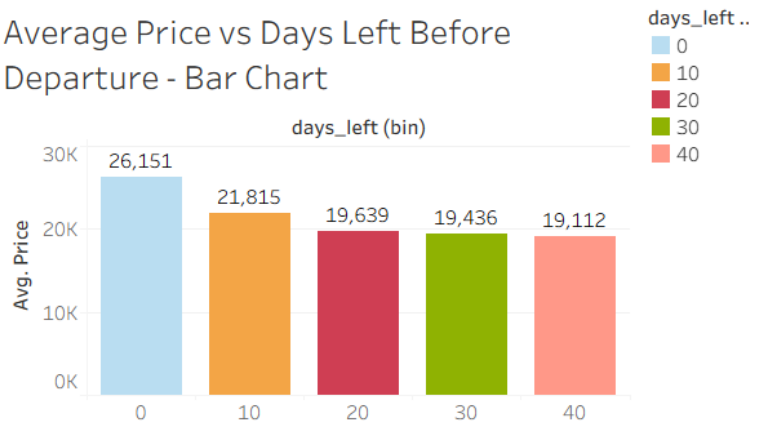
Average Flight Price by 6 Airlines - Bar Chart



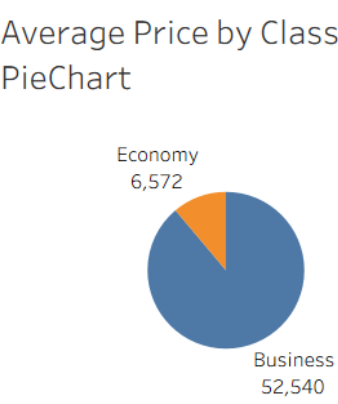
Average Price by Number of Stops - Bar Chart



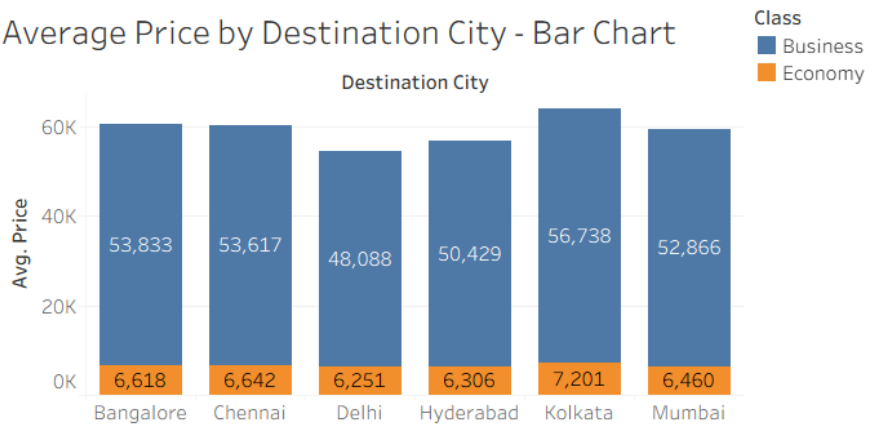
Average Price vs Days Left Before Departure - Bar Chart



Average Price by Class - PieChart



Average Price by Destination City - Bar Chart



INSIGHT ONE – AVERAGE PRICE BY AIRLINES

- Expensive Airlines: Vistara (\$30,397)
 - Premium Air Travel
 - Features Full Services
 - First Indian Carrier to Introduce “Premium Economy Class” (Chandra, November 2024)
- Cheapest Airlines: Air Asia (\$4091)
 - Comes with No Frills (Budget Airlines)
 - Fleet Standardization
 - High-Efficiency Operations
 - Unbundled Fares



INSIGHT TWO – AVERAGE PRICE BY DAYS BEFORE DEPARTURE

- Expensive price if booking 0 days before departure (Last minutes before departure)
 - Reduce the number of available seats
 - Steady demand
 - Combined with limited supply
 - Main Reason: The Supply and Demand Factor
- Cheaper price if booking over 40 days before departure (Over one month before departure)



INSIGHT THREE – AVERAGE PRICE BY NUMBER OF STOPS

- Expensive Price if booked with one stop
 - Higher airport fees
 - Longer travel time
 - Less competition
- Cheaper Price if booked with non-stops
 - Multiple airlines compete and flies with popular non-stop route
 - Demand for the route (May have bigger aircraft with more seats available)
 - Timing (Leaves at less popular time (Early Bird or Red Eye Flight))





INSIGHT FOUR – AVERAGE PRICE BY CLASS

- Expensive Price = Business Class
 - High Standards Service
 - Gourmet Food Impression
 - Digital Expectation of Premium Travel
 - Lounge Access in Airport
- Cheapest Price = Economy Class
 - Basic Standards Service
 - Maximize passenger volume
 - High-volume



INSIGHT FIVE – AVERAGE PRICE BY DESTINATIONS

- Expensive Price: Kolkata
 - Peak-season airfare
 - Luxury accommodation choices
 - Premium Taxi Services
 - Cheapest Price: Delhi
 - Reasonable Price Transportation
 - Affordable Accommodation
 - Cheap Food and Dining
 - Reasonable Cost of Attractions
 - Bargaining Culture
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A close-up photograph of a blue and silver ballpoint pen resting diagonally on a document. The document features a bar chart with blue bars of varying heights. The pen's tip is pointing towards the bottom right of the frame.

NEXT STEP OF THIS PROJECT

- Will conduct SAS software to test Statistical and Regression Model. It can be done in Part Two
- Please Stay Tuned for Part Two



SOURCES:

- <https://www.kaggle.com/datasets/shubhambathwal/flight-price-prediction>
 - <https://www.thehindu.com/news/national/vistara-popularised-premium-air-travel-and-the-airlines-rivals-now-embrace-it/article68867958.ece>
 - <https://traveldave.co.uk/why-airasia-is-the-best-budget-airline/>
 - <https://www.youtube.com/watch?v=Lg4uHq56vEc&t=268>
 - <https://dollarflightclub.com/articles/last-minute-flights-cost-saving-tips/#:~:text=Why%20Last%2DMinute%20Flights%20Are,to%20bump%20up%20prices%20significantly.>
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 - [linkedin.com/posts/aman-singh-695bb3b6_expensive-travel-living-activity-7315636200877240320-FuUg#:~:text=3.,followed%20by%20unrest%20and%20retaliation.](https://www.linkedin.com/posts/aman-singh-695bb3b6_expensive-travel-living-activity-7315636200877240320-FuUg#:~:text=3.,followed%20by%20unrest%20and%20retaliation.)
 - [https://www.visitacity.com/en/new-delhi/guides/new-delhi-on-a-budget#:~:text=Delhi%20is%20a%20budget%20traveler's%20dream.%20You,baksheesh%20\(tipping\)%20which%20is%20expected%20for%20everything.](https://www.visitacity.com/en/new-delhi/guides/new-delhi-on-a-budget#:~:text=Delhi%20is%20a%20budget%20traveler's%20dream.%20You,baksheesh%20(tipping)%20which%20is%20expected%20for%20everything.)
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